

Uncountable Tensor Products and Countable Capture

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1. *Let $(A_\gamma)_{\gamma \in \Gamma}$ be a family of unital C^* -algebras. Assume that uncountably many A_γ 's are nontrivial. Let*

$$A = \bigotimes_{\gamma \in \Gamma}^{\min} A_\gamma$$

be the unital minimal infinite tensor product with respect to the unit embeddings. Then A has the countable-capture property. In particular, A fails BCP.

Corollary 1. *For every uncountable set Γ , the unital C^* -algebra*

$$\bigotimes_{\gamma \in \Gamma}^{\min} M_2$$

has countable capture and fails BCP. This algebra is simple, so its primitive ideal space is a point and has a countable π -base.

Idea of the Proof

Every element of a unital infinite tensor product has countable support: it is a norm limit of finite-support tensors, so the union of the approximating finite supports is countable. Thus a countable family of proposed centres lives inside a countable subproduct.

Choose an unused nontrivial factor A_γ . Inside it choose a self-adjoint contraction v whose spectrum contains both -1 and 1 . Then there are states ρ_+ and ρ_- with

$$\rho_+(v) = 1, \quad \rho_-(v) = -1.$$

For a centre b in the old subproduct and any scalar λ , the slice maps $\text{id} \otimes \rho_\pm$ give

$$\|b - \lambda(1 \otimes v)\| \geq \max\{\|b - \lambda 1\|, \|b + \lambda 1\|\} \geq \|b\|.$$

The last inequality is just the triangle inequality applied to $(b - \lambda 1) + (b + \lambda 1) = 2b$. Therefore

$$\text{dist}(b, \mathbb{C}(1 \otimes v)) = \|b\|$$

for every centre b , which is exactly the quotient-line characterization of countable capture from the previous packet.

Preliminaries

We use the countable-capture characterization proved in the packet

$$2311.14257_countable_capture_characterization.$$

For a nonzero normed space X , countable capture is equivalent to the following condition: for every countable $C \subset X$, there exists $u \in S_X$ such that

$$\text{dist}(c, \mathbb{C}u) = \|c\| \quad (c \in C).$$

Lemma 1 (countable support). *Let*

$$A = \bigotimes_{\gamma \in \Gamma}^{\min} A_\gamma$$

be a unital infinite tensor product. For every $a \in A$, there is a countable subset $\Lambda \subseteq \Gamma$ such that

$$a \in A_\Lambda := \bigotimes_{\gamma \in \Lambda}^{\min} A_\gamma$$

under the canonical unit embedding $A_\Lambda \hookrightarrow A$. Consequently every countable subset of A is contained in A_Λ for some countable $\Lambda \subseteq \Gamma$.

Proof. The infinite tensor product is the norm closure of the algebraic union of the finite subproducts A_F , $F \subseteq \Gamma$ finite. Given $a \in A$, choose finite-support tensors a_n with $a_n \rightarrow a$. Let

$$\Lambda = \bigcup_{n=1}^{\infty} \text{supp}(a_n).$$

Then Λ is countable, every a_n belongs to A_Λ , and A_Λ is closed in A . Hence $a \in A_\Lambda$. Applying this to countably many elements and taking the union of the resulting countable sets gives the final assertion. $\square$

Lemma 2 (two-state element in a nontrivial unital algebra). *Let B be a nontrivial unital C^* -algebra. Then there is a self-adjoint element $v \in B$ with $\|v\| = 1$, and states ρ_+, ρ_- on B , such that*

$$\rho_+(v) = 1, \quad \rho_-(v) = -1.$$

Proof. Choose a self-adjoint non-scalar element $h \in B$. Its spectrum contains at least two points. Let

$$m = \min \sigma(h), \quad M = \max \sigma(h),$$

so $m < M$, and define

$$v = \frac{2h - (M + m)1}{M - m}.$$

Then $v = v^*$, $\|v\| = 1$, and $-1, 1 \in \sigma(v)$. The evaluation characters at -1 and 1 on $C^*(v) \cong C(\sigma(v))$ are states; extend them to states ρ_- and ρ_+ on B . $\square$

Lemma 3 (fresh tensor factor distance). *Let B and D be unital C^* -algebras, let $v \in D$ and states $\rho_\pm$ satisfy Lemma 2. For every $b \in B$ and every $\lambda \in \mathbb{C}$,*

$$\|b \otimes 1 - \lambda(1 \otimes v)\|_{B \otimes_{\min} D} \geq \|b\|.$$

Consequently, for $u = 1 \otimes v$,

$$\text{dist}(b \otimes 1, \mathbb{C}u) = \|b\|.$$

Proof. The slice maps $\text{id}_B \otimes \rho_\pm$ are contractive. Hence

$$\begin{aligned} \|b \otimes 1 - \lambda(1 \otimes v)\| &\geq \max\{\|b - \lambda\rho_+(v)1\|, \|b - \lambda\rho_-(v)1\|\} \\ &= \max\{\|b - \lambda 1\|, \|b + \lambda 1\|\}. \end{aligned}$$

By the triangle inequality,

$$2\|b\| = \|(b - \lambda 1) + (b + \lambda 1)\| \leq \|b - \lambda 1\| + \|b + \lambda 1\| \leq 2 \max\{\|b - \lambda 1\|, \|b + \lambda 1\|\}.$$

Thus the displayed norm is at least $\|b\|$ for every λ . Taking the infimum over λ gives

$$\text{dist}(b \otimes 1, \mathbb{C}u) \geq \|b\|.$$

The reverse inequality follows by taking $\lambda = 0$, and $\|u\| = 1$. $\square$

Proof of the Main Theorem

Proof of Theorem 1. Let $C \subset A$ be countable. By Lemma 1, there is a countable $\Lambda \subseteq \Gamma$ such that

$$C \subseteq A_\Lambda.$$

Since uncountably many factors are nontrivial, choose $\gamma_0 \in \Gamma \setminus \Lambda$ with A_{γ_0} nontrivial. Apply Lemma 2 in A_{γ_0} , obtaining $v \in A_{\gamma_0}$ and states $\rho_\pm$ with $\rho_\pm(v) = \pm 1$.

Let $u \in A$ be the tensor which is v in the coordinate γ_0 and the unit in every other coordinate. Then $\|u\| = 1$. For each $c \in C$, view c as an element of

$$A_\Lambda \otimes_{\min} A_{\gamma_0}$$

with trivial γ_0 -coordinate. Lemma 3 gives

$$\text{dist}(c, \mathbb{C}u) = \|c\|.$$

The countable-capture characterization now implies that A has countable capture. The previous countable-capture obstruction packet then implies that A fails BCP. $\square$

Proof of Corollary 1. Apply Theorem 1 to $A_\gamma = M_2$. The resulting algebra is the uncountable CAR/UHF tensor product. It is the direct limit of the finite matrix algebras $\bigotimes_{\gamma \in F} M_2$, over finite $F \subseteq \Gamma$, with unital full connecting maps; hence it is simple. Therefore its primitive ideal space is a single point, which has a countable π -base. $\square$

Scope and Remaining Wall

This theorem supplies a new internal-size obstruction in the countable primitive- π -base region. It is independent of primitive topology: the uncountable M_2 -tensor product is simple, yet fails BCP by countable capture.

The result does not classify all simple nonseparable C^* -algebras. It applies when the algebra has a literal uncountable tensor-coordinate structure, or when a future argument can reduce a countable centre family to a separable subalgebra with a fresh two-state unit available in a commuting or tensor-independent direction.

Novelty and Search Bounds

The local run indexes were searched for tensor products, infinite tensor products, UHF algebras, countable support, and arXiv:2311.14257. Web searches on June 17, 2026 for BCP with infinite tensor products, UHF algebras, and uncountable tensor products found no exact statement of this theorem. The argument is an agent-produced application of the countable-capture characterization.

Human Review Focus

Reviewers should check:

1. the countable-support lemma for unital infinite tensor products;
2. the construction of v and the two endpoint states in a nontrivial unital C^* -algebra;
3. the slice-map distance inequality;
4. the simplicity assertion in the M_2 -tensor corollary.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Gerard J. Murphy, *C^* -Algebras and Operator Theory*, Academic Press, 1990.
- [3] Run `fa.banach_001`, lane 19, *When the Countable-Capture BCP Obstruction Exists*, solution packet `2311.14257_countable_capture_characterization`, 2026.
- [4] Run `fa.banach_001`, lane 19, *A Countable-Capture Obstruction for the Ball-Covering Property*, solution packet `2311.14257_countable_capture_bcp_obstruction`, 2026.